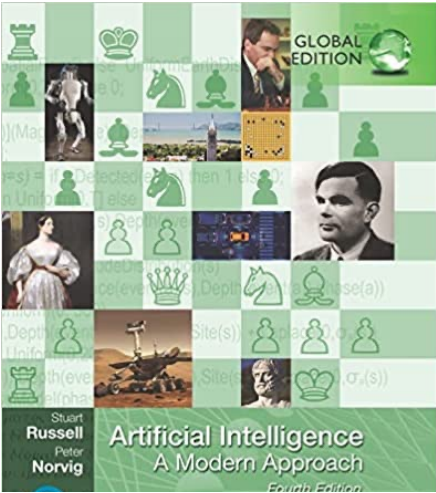


# INTRODUCTION

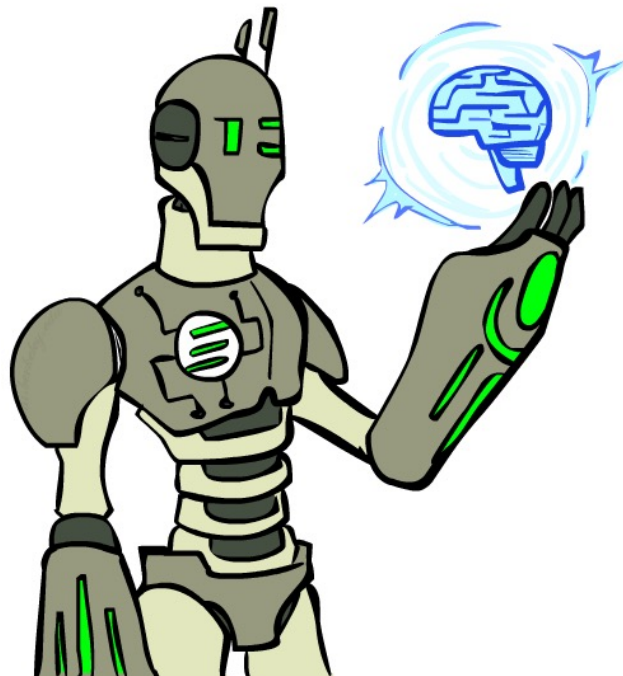


# Slides by Stuart Russell

# Today

---

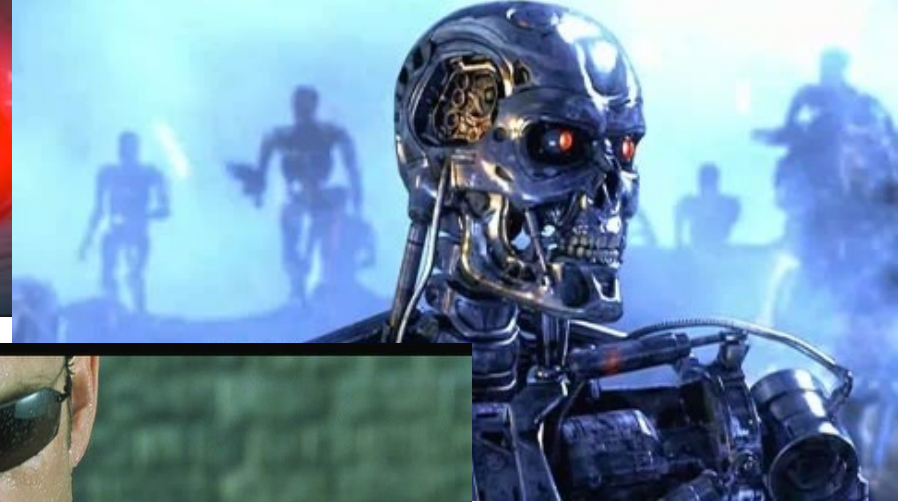
- What is artificial intelligence?
- Where are we and how did we get here?
- How do we think about the design of AI systems?



# Movie AI



# Movie AI





YESTERDAY DR. WILL CASTER WAS ONLY HUMAN

JOHNNY DEPP REBECCA HALL PAUL BETTANY KATE MARA CILLIAN MURPHY AND MORGAN FREEMAN

# TRANSCENDENCE

ALCON ENTERTAINMENT PRESENTS A STRAIGHT UP FILMS PRODUCTION A FILM BY WALLY PFISTER "TRANSCENDENCE" JOHNNY DEPP MORGAN FREEMAN REBECCA HALL KATE MARA CILLIAN MURPHY COLLE HANSEN AND PAUL BETTANY  
CASTING BY MICHAEL DANKO COSTUME DESIGNER STEVE ANDERSON EDITOR GEMINI L. LITTLE EXECUTIVE PRODUCERS JESS HALL AND JESS HALL PRODUCED BY TOLANDA T. COCHRAN STEVEN P. WEISNER BASENBY BONES WRITTEN BY CHRISTOPHER NOLAN EMMA THOMAS DAN KANINZ  
DIRECTED BY WALLY PFISTER  
PRODUCED BY ALCON ENTERTAINMENT  
CASTING BY MICHAEL DANKO COSTUME DESIGNER STEVE ANDERSON EDITOR GEMINI L. LITTLE EXECUTIVE PRODUCERS JESS HALL AND JESS HALL PRODUCED BY TOLANDA T. COCHRAN STEVEN P. WEISNER BASENBY BONES WRITTEN BY CHRISTOPHER NOLAN EMMA THOMAS DAN KANINZ  
DIRECTED BY WALLY PFISTER

entertainmentfilms.co.uk

# News AI

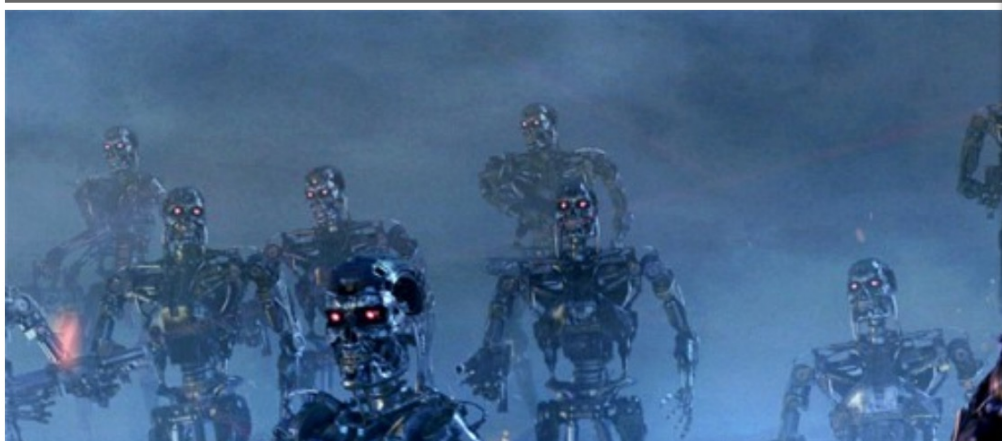
AI is the biggest risk we face as a civilisation, Elon Musk says

## Billionaire burn: Musk says Zuckerberg's understanding of AI threat is limited

HOME » FINANCE » FINANCE TOPICS » DAVOS

'Sociopathic' robots could overrun the human race in a generation

Computers should be trained to serve humans to reduce their threat to the human race, says a leading expert on artificial intelligence



## LIVESCIENCE

NEWS TECH HEALTH PLANET EARTH

Live Science > Tech

## Lifelike 'Sophia' Robot Granted Citizenship to Saudi Arabia

By Mindy Weisberger, Senior Writer | October 30, 2017 03:39pm ET

f 0

0

f

0

0

0

0

0

0

0

0

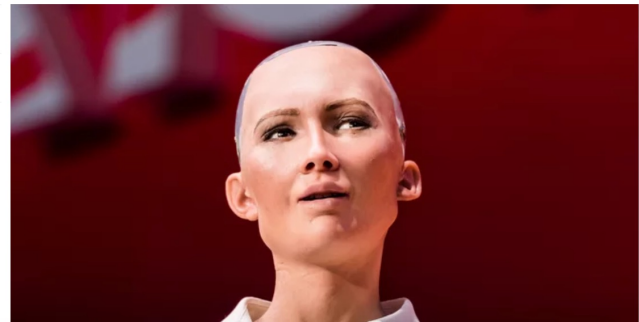
0

0

0

Be the most interesting person you know, subscribe to LiveScience.

Subscribe >



# News AI

TECH • ARTIFICIAL INTELLIGENCE

## United Kingdom Plans \$1.3 Billion Intelligence Push

France to spend \$1.8 billion on compete with U.S., China

EU wants to invest £18b development

## China's Got a Huge Artificial Intelligence Plan

### 'Whoever leads in AI will rule the world': Putin to Russian children on Knowledge Day

Published time: 1 Sep, 2017 14:08

Edited time: 1 Sep, 2017 14:40





# News AI

NATURAL 'PROZAC': DOES IT REALLY WORK?

## IBM's Watson Jeopardy Computer Shuts Down Humans in Final Game

DAILY NEWS 9 March 2016

Sili

### 'I'm in shock!' How world's best hum



Google DeepMind  
Challenge Match  
8 - 15 March 2016

Who is Stoker?  
(I FOR ONE WELCOME OUR  
NEW COMPUTER OVERLORDS)

## Blizzard will show off Google's Deepmind AI in StarCraft 2 later this week

By Andy Chalk 4 hours ago

Google and Blizzard launched the artificial intelligence project in 2016.



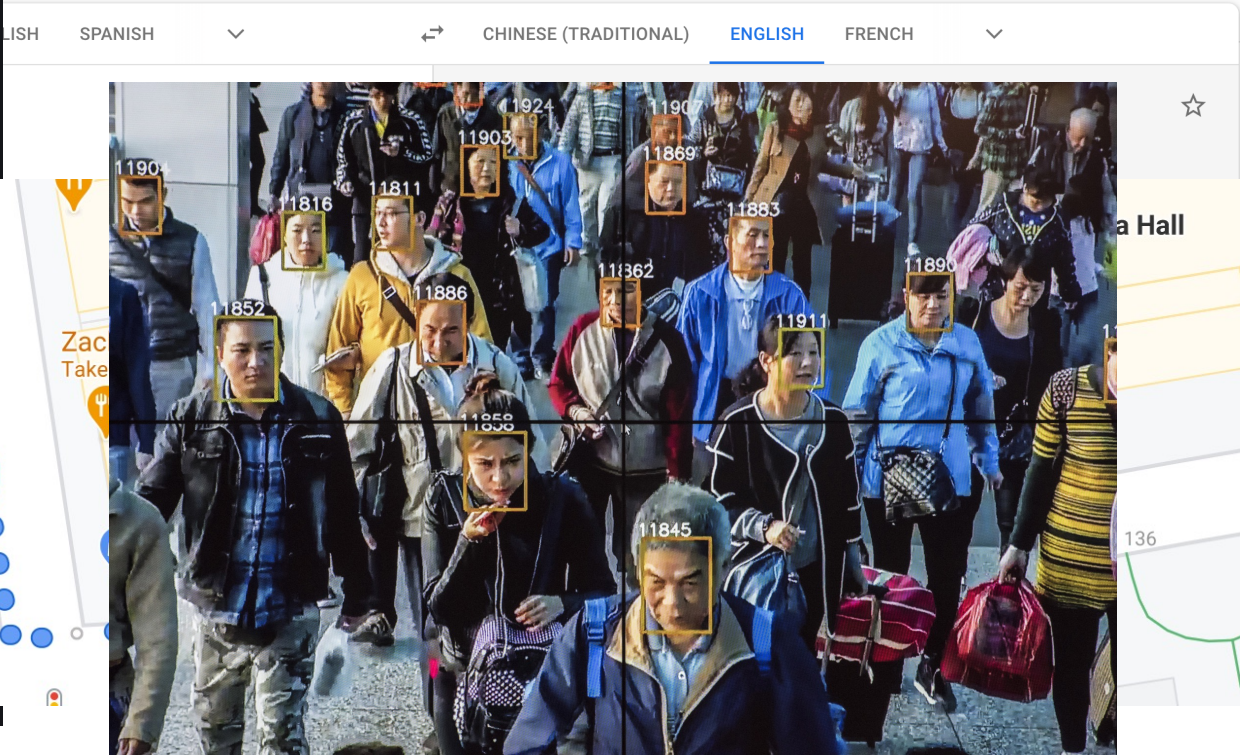
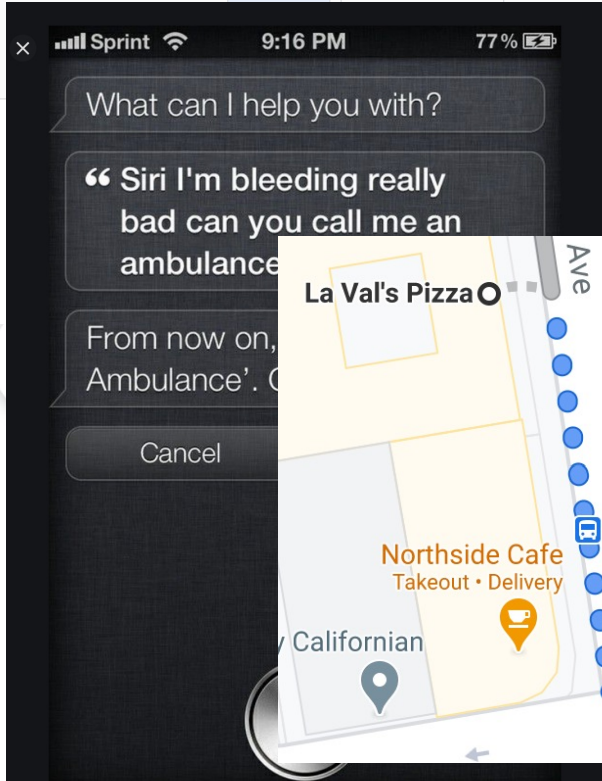
COMMENTS





# Real AI

Google Translate





TUG  
CAUTION  
MAY CONTAIN  
CHEMOTHERAPY DRUG

CAUTION  
MAY CONTAIN  
CHEMOTHERAPY DRUG







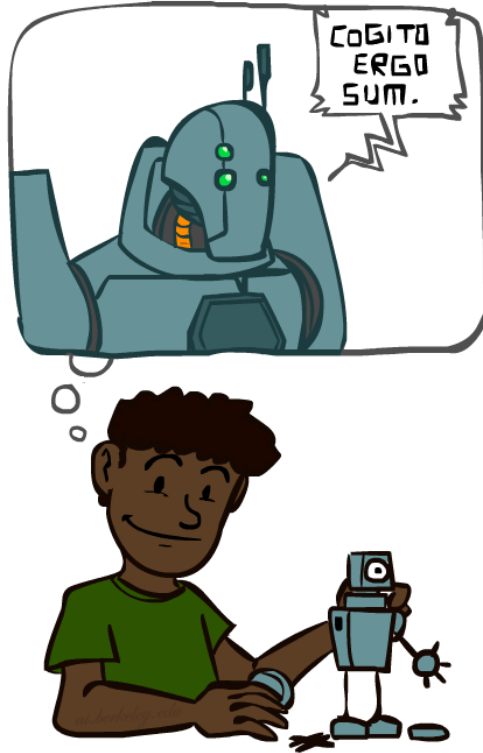


Boston Dynamics



# A (Short) History of AI

---





# Origins of AI

- Influences:

- **Philosophy** (reasoning, planning, learning, science, automation)
- **Mathematics** (logic, probability, optimization)
- Aristotle: For if every instrument could accomplish its own work, obeying or anticipating the will of others . . . if, in like manner, the shuttle would weave and the plectrum touch the lyre without a hand to guide them, chief workmen would not want servants, nor masters slaves
- **Linguistics** (grammars, formal representation of meaning)

- Prehistory:

- Leibniz “Calculus!” (early 1700)
- Babbage Difference Engine (and Analytical Engine (first general-purpose computer, mechanical)
- Lovelace: “a thinking machine” for “all subjects in the universe.”